20.2.3 Plant Breeding for Insects / Pests Resistance

In addition to microorganisms, a large number of insects and pests also cause damage to the crops. Hence, insect and pest resistant crop varieties were developed. Some of them are given below:

Table 20.2 Insects /pests resistant varieties

Crop	Variety	Resistant to Insects/Pests
Brassica	Pusa Gaurav	Aphids
Flat Bean	Pusa Sem 2, Pusa Sem 3	Leaf hopper, aphids and fruit borer
Lady's finger	Pusa Sawani, Pusa A4	Shoot and fruit borer

20.2.4 Plant Breeding for Improved Nutritional Quality

Undernutrition and protein malnutrition among human population is a major health problem which has been receiving much focus throughout the world. Apart, from humans it also affects the health of farm animals. To combat these conditions, human and animal health are to be determined by the nutritional quality of the feed crops. The nutritional quality of crops depends on quality and quantity of nutrients.The nutritional quality may be improved with respect to its

1. Protein content and quality of protein
2. Oil content
3. Mineral content

Biofortification

Biofortification is the scientific process of developing crop plants enriched with high levels of desirable nutrients like vitamins, proteins and minerals. Some examples of crop varieties developed as a result of biofortification are given below:

1. Protina, Shakti and Rathna are lysine rich maize hybrids (developed in India).



Figure 20.2 Protina–lysine rich Maize

2. Atlas 66, a protein rich wheat variety.



Figure 20.3 Atlas 66-protein rich Wheat

3. Iron rich fortified rice variety.
4. Vitamin A enriched carrots, pumpkin and spinach.

20.3 Methods of Plant Breeding for Crop Improvement

Methods of plant breeding to develop high yielding varieties are given below:

1. Introduction of new varieties of plants
2. Selection
3. Polyploidy breeding
4. Mutation breeding
5. Hybridization

20.3.1 Introduction of New Varieties of Plants

It is a process of introducing high yielding varieties of plants from one place to another. Such plants are called as **exotic species**. These imported plant materials may carry pathogens and pests, hence they are thoroughly tested in a plant quarantine before being introduced to the fields. e.g *Phaseolus mungo* was introduced from China.

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